

MTH303 - Endsem

The FL Axioms:

1. $p \rightarrow q \rightarrow p$
2. $[p \rightarrow (q \rightarrow r)] \rightarrow [(p \rightarrow q) \rightarrow (p \rightarrow r)]$
3. $\neg q \rightarrow \neg p \rightarrow (p \rightarrow q)$

1. Consider the propositional form $(p \wedge q) \rightarrow (\neg p \rightarrow r)$
 - (a) (2 points) Is it a tautology?
 - (b) (3 points) Give a formal proof of it using the FL Axioms.
2. Let $p_0, p_1, p_2, \dots$ denote a sequence of propositional forms.
 - (a) (3 points) Prove that if $p_0, p_1, \dots \not\models q$, then, $\neg q, p_0, p_1, \dots$ is consistent.
 - (b) (2 points) Now assume that the sequence is maximally consistent. Prove that $p \wedge q = p_k$ for some k if and only if $p = p_i$ for some i and $q = p_j$ for some j .
3. (3 points) Prove that if $\vdash (\alpha \leftrightarrow \beta)$ then, $\alpha \models \beta$ and $\beta \models \alpha$. (Context: $\models$ is logical implication and $\vdash$ refers to deductions in first order logic.)
4. (5 points) State formally the property that $=$ is symmetric in first order logic. Also give a formal proof of it.
5. (5 points) Consider the well formed formula in ZFC, $\varphi = "\forall x \exists y \neg (y \in x)"$. Is it a sentence? Let Γ denote the ZFC Axioms. Is it true that $\Gamma \vdash \varphi$? Justify it completely.
6. (5 points) Given two sets A and B , define the intersection $A \cap B$ and prove that it is a set. Use only the ZFC axioms, and prove any intermediate assertions you make.
7. (5 points) Given a set relation $R \subset X \times Y$ where X and Y are sets. Use Zorn's Lemma to prove that there exists a function $f \subset R$ so that the domain of f is the domain of R .
8. Recall that ω is defined as the intersection of all inductive subsets of a given inductive set.
 - (a) (2 points) Prove that $n \neq n^+$ for any $n \in \omega$.
 - (b) (3 points) Prove that for any $n \neq \emptyset$, there exists an $m \in \omega$ such that $n = m^+$. Recall that $n^+ = n \cup \{n\}$.
 - (c) Prove that for any $n \in \omega$, if $k \in n$, then, $k \in \omega$